NOTES ON DERIVED COMPLETION AND DUALITY

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1. Derived torsion and completion

Let A be a ring, let $\underline{f} = (f_1, \dots, f_n)$. Write

$$C^{\underline{f}} = \left[A \longrightarrow \prod_{i=1}^{n} A_{f_i} \longrightarrow \prod_{i_1<i_2} A_{f_{i_1} f_{i_2}} \longrightarrow \cdots \longrightarrow A_{f_1 \cdots f_n} \right] \simeq C^{f_1} \otimes_A^L \cdots \otimes_A^L C^{f_n},$$

where A sits at cohomological degree 0.

Lemma 1.1. *$C^{\underline{f}}$ is represented by a complex of projetive A-modules in the range $[0, n]$.*

Proof. We may assume $n = 1$ and it suffices to show $\mathrm{pd}_A A_f \le 1$ for all $f \in A$. This is true because we have the explicit resolution

$$0 \longrightarrow \oplus_{\mathbf{N}} A \longrightarrow \oplus_{\mathbf{N}} A \longrightarrow A_f \longrightarrow 0$$

where the first map sends $(a_0, a_1, \dots)$ to $(a_0, a_1 - f a_0, a_2 - f a_1 \dots)$ and the second map sends $(a_0, a_1, \dots)$ to $a_0 + a_1/f + a_2/f^2 + \dots$. □

There is a distinguished triangle

$$C^{\underline{f}} \longrightarrow A \longrightarrow (A_h\text{-complexes}, h \in (\underline{f})) \longrightarrow +1. \tag{1}$$

For one element f, this is

$$C^f \longrightarrow A \longrightarrow A_f \longrightarrow +1.$$

For $X \in \mathrm{D}(A)$, we have the natural maps

$$R\Gamma_{\underline{f}} X := X \otimes_A^L C^{\underline{f}} \longrightarrow X, \qquad X \longrightarrow \widehat{X}^{\underline{f}} := \mathrm{RHom}_A(C^{\underline{f}}, X).$$

Without any understanding of the notions yet, we already have our first theorem via tensor-hom adjunction.

Theorem 1.2. *For $X, Y \in \mathrm{D}(A)$,*

$$\mathrm{RHom}_A(R\Gamma_{\underline{f}} X, Y) = \mathrm{RHom}_A(X, \widehat{Y}^{\underline{f}}).$$

Definition 1.3. Let $f \in A$. An object $X \in \mathrm{D}(A)$ is *f^∞-torsion* if $X \otimes_A^L A_f = 0$, or equivalently if $R\Gamma_f X \xrightarrow{\sim} X$. It is *$f$-complete* if $\mathrm{RHom}_A(A_f, X) = 0$, or equivalently if $X \xrightarrow{\sim} \widehat{X}^f$. In the torsion condition, one may equivalently require $X \otimes_A^L Z = 0$ for every A_f-complex Z: indeed, this tensor product is $(X \otimes_A^L A_f) \otimes_{A_f}^L Z$. In the completeness condition, one may equivalently require $\mathrm{RHom}_A(Z, X) = 0$ for every A_f-complex Z, by the adjunction

$$\mathrm{RHom}_A(Z, X) = \mathrm{RHom}_{A_f}(Z, \mathrm{RHom}_A(A_f, X)).$$

Theorem 1.4 ([Stacks, Tag 091Q])**.** *Let $X \in \mathrm{D}(A)$. The sets*

$$\{f \in A \mid X \text{ is } f^\infty\text{-torsion}\}, \qquad \{f \in A \mid X \text{ is } f\text{-complete}\}$$

are radical ideals.

If these ideals contain an ideal I, we say that X is *I^∞-torsion* and *I-complete*, respectively.

Proof. To see the sets are closed under sums, we have a distinguished triangle

$$A_{f+g} \longrightarrow A_{f(f+g)} \oplus A_{g(f+g)} \longrightarrow A_{fg(f+g)} \longrightarrow +1.$$

Indeed, in A_{f+g} the elements f and g generate the unit ideal. Tensoring this triangle with X proves closure under addition for the torsion condition. Applying $\mathrm{RHom}_A(-, X)$ proves closure under addition for the completeness condition. Similarly but more easily, we get that the sets are closed under products. □

Theorem 1.5. *Let $I = (\underline{f})$. The inclusions of the thick full subcategories*

$$\mathrm{D}_{I\text{-comp}}(A) \subseteq \mathrm{D}(A) \supseteq \mathrm{D}_{I^\infty\text{-tors}}(A)$$

have adjoints

$$\widehat{(-)}^{\underline{f}} \dashv (\mathrm{D}_{I\text{-comp}}(A) \hookrightarrow \mathrm{D}(A)), \qquad (\mathrm{D}_{I^\infty\text{-tors}}(A) \hookrightarrow \mathrm{D}(A)) \dashv R\Gamma_{\underline{f}}(-).$$

In particular, $\widehat{(-)}^{\underline{f}}$ and $R\Gamma_{\underline{f}}(-)$ do not depend on $\underline{f}$, but only on $\sqrt{I}$. We will write $\widehat{(-)}^{I}$ and $R\Gamma_I(-)$ instead.

We use

Lemma 1.6. *For every A_f-complex X, $\mathrm{RHom}_A(A_f, X) = X$; alternatively, $\widehat{X}^f = 0$.*

Proof. Since $\mathrm{pd}_A A_f \leq 1$, we may assume $X \in \mathrm{D}^+(A)$. Then X is represented by a complex of injective A-modules that are also A_f-modules: the injective hull of every f^∞-torsion-free module is f^∞-torsion-free, hence an A_f-module. Now the result follows from $\mathrm{Hom}_A(A_f, M) = M$ for every $M \in \mathrm{D}^\heartsuit(A_f)$. □

Proof of Theorem 1.5. Let $Y \in \mathrm{D}_{I\text{-comp}}(A)$ and $X \in \mathrm{D}(A)$. The map $X \to \widehat{X}$ induces

$$\mathrm{RHom}_A(\widehat{X}, Y) \longrightarrow \mathrm{RHom}_A(X, Y).$$

Its fiber is of the form $\mathrm{RHom}_A(A_f\text{-complexes}, Y)$ where $f \in I$ by (1), hence zero.

Let $X \in \mathrm{D}_{I^\infty\text{-tors}}(A)$ and $Y \in \mathrm{D}(A)$. The natural map induces

$$\mathrm{RHom}_A(X, R\Gamma_{\underline{f}} Y) \longrightarrow \mathrm{RHom}_A(X, Y).$$

Its cofiber is of the form $\mathrm{RHom}_A(X, A_f\text{-complexes})$ where $f \in I$ by (1). For $f \in I$ and Z an A_f-complexes, by Lemma 1.6 and adjunction,

$$\begin{aligned}\mathrm{RHom}_A(X, Z) &= \mathrm{RHom}_A(X, \mathrm{RHom}_A(A_f, Z)) \\ &= \mathrm{RHom}_A(X \otimes^L_A A_f, Z) = 0,\end{aligned}$$

as $X \otimes^L_A A_f = 0$ □

Lemma 1.7. *The canonical map*

$$C^{\underline{f}} \otimes^L_A C^{\underline{f}} \xrightarrow{\sim} C^{\underline{f}}$$

is an isomorphism.

Proof. Looking at the definition of $C^{\underline{f}}$ we need to show for all i and all $h \in A$ we have $C^{\underline{f}} \otimes^L_A A_{f_i h} = 0$. This is because $C^{f_i} \otimes^L_A A_{f_i h} = [A_{f_i h} \to A_{f_i f_i h}] = 0$. □

Corollary 1.8. *The functors $\widehat{(-)}^{I}$ and $R\Gamma_I(-)$ are idempotent.*

Lemma 1.9. *Omitting the ideal from the completion notation, we have*

$$\widehat{\mathrm{RHom}_A(X, Y)} = \mathrm{RHom}_A(\widehat{X}, \widehat{Y}) = \mathrm{RHom}_A(X, \widehat{Y}).$$

Proof. The second equality follows from Theorem 1.5. For the first,

$$\begin{aligned}\mathrm{RHom}_A(C^{\underline{f}}_-, \mathrm{RHom}_A(X,Y)) &= \mathrm{RHom}_A(C^{\underline{f}}_- \otimes^L_A X, Y)\\ &= \mathrm{RHom}_A(X \otimes^L_A C^{\underline{f}}_-, Y)\\ &= \mathrm{RHom}_A(X, \widehat{Y}),\end{aligned}$$

by Theorem 1.2. □

Theorem 1.10.

$$\mathrm{RHom}_A(\widehat{X}^I, \widehat{Y}^I) = \mathrm{RHom}_A(R\Gamma_I X, R\Gamma_I Y).$$

Proof. By Theorem 1.2, the right side is $\mathrm{RHom}_A(X, \widehat{R\Gamma_I Y})$, whereas the left side is $\mathrm{RHom}_A(X, \widehat{Y})$ by Lemma 1.9. Consider the triangle

$$R\Gamma_I Y \longrightarrow Y \longrightarrow Q \longrightarrow +1.$$

We need $\widehat{Q} = 0$. It suffices to show that $\widehat{Z} = 0$ for every A_f-complex Z, where $f \in I$. This is because we can use any set of generators $f = f_1, \dots, f_n$ of I to compute $\widehat{Z}$, so $\widehat{Z} = \mathrm{RHom}_A(C^{\underline{f}}_-, Z) = \mathrm{RHom}_A(C^{f_2,\dots,f_n}, \mathrm{RHom}_A(C^f, Z)) = 0$ (Lemma 1.6). □

Corollary 1.11. *For every $Y \in \mathrm{D}(A)$, the natural map induces an isomorphism*

$$\widehat{R\Gamma_{\underline{f}} Y} \xrightarrow{\sim} \widehat{Y}.$$

Proof. In the triangle in the preceding proof, $\widehat{Q} = 0$. Applying completion therefore gives the asserted isomorphism. □

Proposition 1.12. *For every $X \in \mathrm{D}(A)$,*

$$\begin{aligned} X \in \mathrm{D}_{\mathrm{tors}}(A) &\iff \mathrm{H}^i(X) \in \mathrm{D}^{\heartsuit}_{\mathrm{tors}}(A) \text{ for every } i,\\ X \in \mathrm{D}_{\mathrm{comp}}(A) &\iff \mathrm{H}^i(X) \in \mathrm{D}^{\heartsuit}_{\mathrm{comp}}(A) \text{ for every } i.\end{aligned}$$

Proof. The first assertion is immediate because tensoring with A_f is flat. For the second, $\mathrm{RHom}_A(A_f, -)$ has cohomological dimension $[-1, 0]$, so we have

$$\begin{aligned}&\mathrm{Ext}^1_A(A_f, \mathrm{H}^{i-1}(X)) = 0 = \mathrm{Hom}_A(A_f, \mathrm{H}^i(X))\\ &\iff \quad \mathrm{H}^i \mathrm{RHom}_A(A_f, X) = 0.\end{aligned}$$

□

Corollary 1.13. $\mathrm{D}^{\heartsuit}_{\mathrm{comp}}(A)$ *is a weak Serre subcategory.*

Proof. As an illustration, let us see why $\mathrm{D}^{\heartsuit}_{\mathrm{comp}}(A)$ is stable under kernels and cokernels. Let $\varphi : M \to N \in \mathrm{D}^{\heartsuit}_{\mathrm{comp}}(A)$, then $[M \to N] \in \mathrm{D}^{\heartsuit}_{\mathrm{comp}}(A)$ as $\mathrm{D}^{\heartsuit}_{\mathrm{comp}}(A)$ is a thick subcategory. Apply Proposition 1.12 we see $\ker(\varphi)$ and $\mathrm{coker}(\varphi)$ are in $\mathrm{D}^{\heartsuit}_{\mathrm{comp}}(A)$. □

Lemma 1.14 (Nakayama)**.** *Let $M \in \mathrm{D}^{\heartsuit}_{I\text{-comp}}(A)$. If $M = IM$, then $M = 0$.*

Proof. Say $I = (f_1, \dots, f_n)$. By Corollary 1.13, $N = (f_2, \dots, f_n)M$ and M/N are in $\mathrm{D}^{\heartsuit}_{\mathrm{comp}}(A)$ (N being the image of a map $M^{\oplus(n-1)} \to M$). We have $M = f_1 M + N$. Induction on n reduces us to $n = 1$. Then $M = fM$, and, if $M \neq 0$, elementary considerations give $\mathrm{Hom}_A(A_f, M) \neq 0$. □

Corollary 1.15. *Let $X \in \mathrm{D}_{I\text{-comp}}(A)$ where $I = (f_1, \dots, f_n)$. The following are equivalent for every integer c.*

(i) $X \in \mathrm{D}^{\leq c}$.
(ii) $X \otimes^L_A A/I \in \mathrm{D}^{\leq c}$.
(iii) $X \otimes^L_A K \in \mathrm{D}^{\leq c}$, *where $K = \mathrm{Kos}(f_1, \dots, f_n)$ is the Koszul complex.*

Proof. Trivially (i) implies (ii), and (ii) implies (iii) as $X \otimes^L_A K = X \otimes^L_A A/I \otimes^L_{A/I} K$. Now assume (iii) and let $Y = X \otimes^L_A \mathrm{Kos}(f_1, \dots, f_{n-1})$. As $\mathrm{Kos}(f_1, \dots, f_{n-1})$ is a perfect complex we have $Y \in \mathrm{D}_{I\text{-comp}}(A)$, and (iii) gives $\mathrm{H}^i(Y) = f_n \mathrm{H}^i(Y)$ for all $i > 0$. Proposition 1.12 and Lemma 1.14 give $\mathrm{H}^i(Y) = 0$ for all $i > 0$, and we win by induction. □

Lemma 1.16 ([Lyu25, Lemma 5.4])**.** *Let $X \in \mathrm{D}_{I\text{-comp}}(A)$, $f \in I$, and*

$$C := \mathrm{cofib}(X \xrightarrow{\times f} X).$$

(i) *If $C = 0$, then $X = 0$.*
(ii) *Assume A is Noetherian. If $A \in \mathrm{D}_{I\text{-comp}}(A)$ and $X \in \mathrm{D}^-(A)$, then $C \in \mathrm{D}^-_{\mathrm{coh}}(A)$ implies $X \in \mathrm{D}^-_{\mathrm{coh}}(A)$.*

Proof. We get (i) trivially from Corollary 1.15. To prove (ii), fix $c \in \mathbf{Z}$ and let us show $\mathrm{H}^c(X)$ is finitely generated.

Let $b \in \mathbf{Z}$ be such that $X \in \mathrm{D}^{\leq b}$ and not in $\mathrm{D}^{\leq b-1}$. We use induction on b, as when $b < c$ the result is trivial. We have $\mathrm{H}^b(X) \in \mathrm{D}_{I\text{-comp}}(A)$ and $\mathrm{H}^b(X)/f\mathrm{H}^b(X)$ is finitely generated. Therefore there exists a map $\varphi : F \to \mathrm{H}^b(X)$ inducing a surjective map on $(-)/f(-)$, where F is a finite free A-module. Then φ is surjective by Lemma 1.14. The map φ comes from a map $\psi : F[-b] \to X$, and as $A \in \mathrm{D}_{I\text{-comp}}(A)$, we have $\mathrm{cofib}(\psi) \in \mathrm{D}_{I\text{-comp}}(A)$, which has smaller b, we win. □

2. Finitely genereted modules over Noetherian rings

Theorem 2.1 (Derived and classical completions coincide for finite modules over Noetherian rings)**.** *Let A be a Noetherian ring and I and ideal of A. Then for every $M \in \mathrm{D}^\heartsuit_{\mathrm{coh}}(A)$ we have $\widehat{M}^I = \lim_t M/I^t M$.*

[To be built in the future. This follows from [Stacks, Tag 0A06]; we want to supplement an interesting elementary argument that avoids derived limits.]

3. Matlis duality

In the next three sections, let $(A, \mathfrak{m}, k)$ be a Noetherian complete local ring (in particular, $\mathfrak{m}$ is finitely generated). We omit $\mathfrak{m}$ from the notation when it is obvious (e.g. $\mathrm{D}_{\mathrm{comp}}$ means $\mathrm{D}_{\mathfrak{m}\text{-comp}}$).

We give the axiomatic description of the injective hull of k (cf. [Stacks, Tag 08Y3]). We will see such an object is unique (Prop 5.2).

Theorem 3.1. *There exists $E \in \mathrm{D}^\heartsuit(A)$ such that*

(i) *E is injective;*
(ii) *$\mathrm{Hom}_A(k, E) \simeq k$;*
(iii) *E is $\mathfrak{m}^\infty$-torsion (cf. [Stacks, Tag 08XW]).*

Note that (i) and (ii) imply $\mathrm{RHom}_A(k, E) \simeq k$.

Define

$$(-)^\vee := \mathrm{RHom}_A(-, E)$$

so for all $M \in \mathrm{D}(A)$ we have a canonical map $\mu_M : M \longrightarrow M^{\vee\vee}$.

Remark 3.2. The canonical map $\mu_k : k \to k^{\vee\vee}$ is an isomorphism. Indeed, $\mu_M : M \longrightarrow M^{\vee\vee}$ is nonzero for any $M \in \mathrm{D}^\heartsuit(A)$ as soon as $\mathrm{RHom}_A(M, E) \neq 0$, since $\mu_M(x) = (\varphi \mapsto \varphi(x))$ is the evaluation map. But any nonzero map between two isomorphic copies of k is an isomorphism.

Theorem 3.3. *For any $M \in \mathrm{D}_{\mathrm{coh}}(A)$, the canonical map*

$$\mu_M : M \longrightarrow \mathrm{RHom}_A(\mathrm{RHom}_A(M, E), E)$$

is an isomorphism.

Proof. Since E is injective, we may assume M is bounded. Induct on $d = \dim \operatorname{Supp} M$. If $d = 0$, M is a successive extension of shifts of k, so we win by Remark 3.2. Assume $d > 0$ and the theorem is true fo smaller d, take $f \in \mathfrak{m}$ such that

$$\dim(V(f) \cap \operatorname{Supp} M) < d.$$

Consider the distinguished triangle which defines C

$$M \xrightarrow{\times f} M \longrightarrow C \longrightarrow +1.$$

Applying double duality gives a morphism of distinguished triangles

$$\begin{array}{ccccccc} M & \xrightarrow{\times f} & M & \longrightarrow & C & \longrightarrow & +1 \\ \downarrow{\scriptstyle \mu_M} & & \downarrow{\scriptstyle \mu_M} & & \downarrow{\scriptstyle \mu_C} & & \\ M^{\vee\vee} & \xrightarrow{\times f} & M^{\vee\vee} & \longrightarrow & C^{\vee\vee} & \longrightarrow & +1 \end{array}$$

We know $\operatorname{Supp} C \subseteq V(f) \cap \operatorname{Supp} M$. By the induction hypothesis, μ_C is an isomorphism. Thus $\times f$ is an isomorphism on $\operatorname{cofib}(\mu_M)$. However, $M \in \mathrm{D}_{\mathrm{comp}}(A)$ as A is complete, and $M^{\vee\vee} \in \mathrm{D}_{\mathrm{comp}}(A)$: both $\operatorname{RHom}_A(M, E)$ and E are $\mathfrak{m}^\infty$-torsion, as $M \in \mathrm{D}^b_{\mathrm{coh}}(A)$, so Theorem **??** and Lemma **??** tell us $M^{\vee\vee} \in \mathrm{D}_{\mathrm{comp}}(A)$. Hence $\operatorname{cofib}(\mu_M) \in \mathrm{D}_{\mathrm{comp}}(A)$. By Nakayama, $\operatorname{cofib}(\mu_M) = 0$. □

Corollary 3.4. $\operatorname{Hom}_A(E, E) = A$.

Remark 3.5. There is an equivalence of abelian categories (Matlis duality)

$$\{\text{Noetherian (i.e. finitely generated) } A\text{-modules}\} \underset{(-)^\vee}{\overset{(-)^\vee}{\rightleftarrows}} \{\text{Artinian } A\text{-modules}\}.$$

We have proved part of it ($\mathrm{id} = (-)^{\vee\vee}$ on the category on the left hand side). The rest of the proof is left to the interested reader (not trivial).

4. Local duality

Definition 4.1. The normalized dualizing complex is

$$\omega_A^\bullet := \widehat{E},$$

and we set, for every $M \in \mathrm{D}(A)$

$$\mathbb{D}(M) := \operatorname{RHom}_A(M, \omega_A^\bullet) \in \mathrm{D}_{\mathrm{comp}}(A) \text{ (cf. Lemma 1.9).}$$

Again uniqueness will be dealt with the next time.

Theorem 4.2 (The local duality theorem, [Stacks, Tag 0A84])**.** *For $M \in \mathrm{D}(A)$,*

$$\mathbb{D}(M) = \operatorname{RHom}_A(R\Gamma_{\mathfrak{m}} M, E) \in \mathrm{D}_{\mathrm{comp}}(A).$$

In particular, $\mathbb{D}(k) \simeq k$.

Corollary 4.3. *If $M \in \mathrm{D}^{[a,b]}(A)$, then $\mathbb{D}(M) \in \mathrm{D}^{[-b-\dim A, -a]}(A)$. If $M \in \mathrm{D}^{\heartsuit}(A)$, then $\mathbb{D}(M) \in \mathrm{D}^{[-\dim A, 0]}(A)$. (This is because $\mathfrak{m}$ is the radical of an ideal generated by $\dim A$ elements, hence $R\Gamma_{\mathfrak{m}}(-)$ has cohomological dimension $\dim A$.)*

In particular, $\omega_A^\bullet \in \mathrm{D}^{[-\dim A, 0]}(A)$.

Proof of Theorem 4.2.

$$\begin{aligned} \mathbb{D}(M) &= \operatorname{RHom}_A(M, \omega_A^\bullet) \\ &= \operatorname{RHom}_A(\widehat{M}, \widehat{E}) && \text{(Lemma 1.9 and Definition)} \\ &= \operatorname{RHom}_A(R\Gamma_{\mathfrak{m}} M, R\Gamma_{\mathfrak{m}} E) && \text{(Theorem 1.10)} \\ &= \operatorname{RHom}_A(R\Gamma_{\mathfrak{m}} M, E) && (E \in \mathrm{D}_{\mathrm{tors}}(A)). \end{aligned}$$

□

Theorem 4.4. *For every $M \in \mathrm{D}_{\mathrm{coh}}(A)$, the canonical map*

$$\gamma_M : M \longrightarrow \mathbb{D}(\mathbb{D}(M))$$

is an isomorphism.

Proof. The proof is the same as for μ_M. Indeed, by Corollary 4.3, we may assume $M \in \mathrm{D}^b_{\mathrm{coh}}(A)$. Using $\mathbb{D}(\mathbb{D}(M)) \in \mathrm{D}_{\mathrm{comp}}(A)$, we can reduce to the case $M = k$. Theorem 1.10 tells us for $X \in \mathrm{D}_{\mathrm{comp}} \cap \mathrm{D}_{\mathrm{tors}}$ we have $\mathbb{D}(X) = X^\vee$. Therefore

$$\mathbb{D}(\mathbb{D}(k)) = \mathbb{D}(k^\vee) = k^{\vee\vee},$$

and γ_k corresponds to μ_k. □

Theorem 4.5. *If $M \in \mathrm{D}_{\mathrm{coh}}(A)$, then $\mathbb{D}(M) \in \mathrm{D}_{\mathrm{coh}}(A)$ (in particular, $\omega_A^\bullet \in \mathrm{D}_{\mathrm{coh}}(A)$).*

Proof. By Corollary 4.3, it suffices to consider the case $M \in \mathrm{D}^b_{\mathrm{coh}}(A)$. Again, induct on $d = \dim \operatorname{Supp} M$. If $d = 0$, we may assume $M = k$, and $\mathbb{D}(k) \simeq k$. If $d > 0$, take $f \in \mathfrak{m}$ with $\dim(V(f) \cap \operatorname{Supp} M) < d$ and consider the triangle

$$M \xrightarrow{\times f} M \longrightarrow C \longrightarrow +1$$

which defines C. Dualizing gives

$$\mathbb{D}(C) \longrightarrow \mathbb{D}(M) \xrightarrow{\times f} \mathbb{D}(M) \longrightarrow +1.$$

By induction, $\mathbb{D}(C) \in \mathrm{D}^b_{\mathrm{coh}}(A)$, and $\mathbb{D}(M) \in \mathrm{D}^b_{\mathrm{comp}}(A)$ (Theorem 4.2 and Corollary 4.3). By Nakayama, $\mathbb{D}(M) \in \mathrm{D}_{\mathrm{coh}}(A)$. □

5. Uniqueness

Lemma 5.1. *Let $X \in \mathrm{D}^+_{\mathrm{tors}}(A)$. If*

$$\mathrm{RHom}_A(k, X) \in \mathrm{D}^{\geq \alpha}(A),$$

then $X \in \mathrm{D}^{\geq \alpha}(A)$.

Proof. Let β be such that $X \in \mathrm{D}^{\geq \beta}$ and $X \notin \mathrm{D}^{\geq \beta+1}$. Then

$$\mathrm{H}^\beta \mathrm{RHom}_A(k, X) \simeq \mathrm{Hom}_A(k, \mathrm{H}^\beta(X)) \neq 0.$$

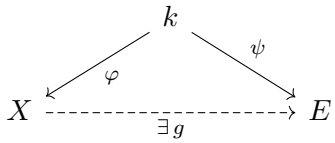
Thus $\beta \geq \alpha$. □

The more general form of Lemma 5.1 is dual to Corollary 1.15 and is left as an exercise.

Proposition 5.2. *Let $X \in \mathrm{D}^+_{\mathrm{tors}}(A)$ be such that $\mathrm{RHom}_A(k, X) \simeq k$. Then $X \simeq E$.*

Proof. The proof proceeds as follows.

(1) $X \in \mathrm{D}^{\geq 0}$ (Lemma 5.1).
(2) Let $\varphi : k \to X$ be a generator of $\mathrm{Hom}_{\mathrm{D}(A)}(k, X) = \mathrm{H}^0(\mathrm{RHom}_A(k, X)) \simeq k$, and $\psi : k \to E$ be a generator of $\mathrm{Hom}_{\mathrm{D}(A)}(k, E) = \mathrm{H}^0(\mathrm{RHom}_A(k, E)) \simeq k$, Set $C = \mathrm{cofib}(\varphi)$. Then $C \in \mathrm{D}^{\geq 0}(A)$, so $\mathrm{RHom}_A(C, E) \in \mathrm{D}^{\leq 0}(A)$, therefore in the following diagram g exists.



(3) By choice, both φ and ψ induce an isomorphism on $\mathrm{Hom}_{\mathrm{D}(A)}(k,-)$, thus by assumption, g induces an isomorphism on $\mathrm{RHom}_A(k,-)$. Hence

$$\mathrm{RHom}_A(k,\mathrm{cofib}(g))=0,$$

which implies $\mathrm{cofib}(g)=0$ (Lemma 5.1).

Consequently $X\simeq E$. □

Proposition 5.3. *Let $Y\in \mathrm{D}^+_{\mathrm{comp}}(A)$ be such that $\mathrm{RHom}_A(k,Y)\simeq k$. Then $Y\simeq \omega_A^\bullet$.*

Proof. We have

$$\mathrm{RHom}_A(k,\mathrm{R}\,\Gamma_{\mathfrak{m}}Y)\simeq k,$$

so $\mathrm{R}\,\Gamma_{\mathfrak{m}}Y\simeq E$ by the preceding proposition. Hence

$$Y\simeq \widehat{\mathrm{R}\,\Gamma_{\mathfrak{m}}Y}\simeq \widehat{E}=\omega_A^\bullet.$$

□

6. Dualizing complexes: the standard definition

In the next three sections A denotes an arbitrary Noetherian ring.

Definition 6.1. An object $\omega^\bullet\in \mathrm{D}(A)$ is a *dualizing complex* if

(1) $\omega^\bullet$ has finite injective dimension, i.e. there exists $\alpha,\beta\in\mathbb{Z}$ such that for all $X\in \mathrm{D}^\heartsuit(A)$,

$$\mathrm{RHom}_A(X,\omega^\bullet)\in \mathrm{D}^{[\alpha,\beta]}(A);$$

this is also equivalent to that $\omega^\bullet$ is represented by a bounded complex of injectives.

(2) $\omega^\bullet\in \mathrm{D}_{\mathrm{coh}}(A)$.

(3) the natural map is an isomorphism:

$$A=\mathrm{RHom}_A(\omega^\bullet,\omega^\bullet).$$

Note that dualizing complexes are stable under localization [Stacks, Tag 0A7G]. Note also that the normalized dualizing complex $\omega_A^\bullet$ for a complete local A considered previously is a dualizing complex, see Corollary 4.3, Theorem 4.5, and Theorem 4.4.

Theorem 6.2. *If A is local, then*

$$\omega^\bullet\otimes^L_A\widehat{A}\simeq \omega^\bullet_{\widehat{A}}[\delta]$$

for some $\delta\in\mathbf{Z}$.

We say $\omega^\bullet$ is *normalized* if $\delta=0$.

Proof. It suffices to show (by Proposition 5.3 and the fact that RHom commutes with flat base change if the source in pseudo-coherent)

$$\mathrm{RHom}_A(k,\omega^\bullet)\simeq k[\delta].$$

From conditions (1) and (3), for $M\in \mathrm{D}^{\mathrm{b}}_{\mathrm{coh}}(A)$,

$$M=\mathrm{RHom}_A(\mathrm{RHom}_A(M,\omega^\bullet),\omega^\bullet).$$

In particular,

$$k=\mathrm{RHom}_A(\mathrm{RHom}_A(k,\omega^\bullet),\omega^\bullet).$$

By (1) and (2), $\mathrm{RHom}_A(k,\omega^\bullet)\in \mathrm{D}^{\mathrm{b}}_{\mathrm{coh}}(k)$, hence is of the form $\bigoplus_{i=1}^t k[a_i]^{\oplus d_i}$. Then

$$k=\mathrm{RHom}_A(\mathrm{RHom}_A(k,\omega^\bullet),\omega^\bullet)\cong\bigoplus_{j=1}^t\left(\bigoplus_{i=1}^t k[a_i]^{\oplus d_i}\right)[a_j]^{\oplus d_j},.$$

forcing $t=1=d_1$; and we put $\delta=a_1$. □

Corollary 6.3. *Let $X, Y \in \mathrm{D}(A)$ be two dualizing complexes. Then the object $\mathcal{L} := \mathrm{RHom}_A(X, Y)$ is invertible*[1] *and the canonical map $X \otimes^L_A \mathcal{L} \to Y$ is an isomorphism.*

In particular, two dualizing complexes of a Noetherian local ring differ only by a shift, and two normalized dualizing complexes of a Noetherian local ring are isomorphic.

Proof. By definition we have $X, Y, \mathcal{L} \in \mathrm{D}^{\mathrm{b}}_{\mathrm{coh}}(A)$, hence the statement is local, so we may assume A is local. After the faithfully flat base change $A \to \widehat{A}$, both X and Y are shifts of $\omega^\bullet_{\widehat{A}}$, and our result follows from part (3) of Definition 6.1. □

Remark 6.4. Not all Noetherian rings have dualizing complexes.

7. Properties of dualizing complexes

Lemma 7.1. *Let $A \to B$ be a finite ring map, and let $\omega^\bullet$ be a dualizing complex of A. Then*

$$\omega^\bullet_B := \mathrm{RHom}_A(B, \omega^\bullet)$$

is a dualizing complex of B.

If both A and B are local and $\omega^\bullet$ is normalized, then $\omega^\bullet_B$ is normalized.

Proof. For $N \in \mathrm{D}(B)$, adjunction gives

$$\mathrm{RHom}_B(N, \mathrm{RHom}_A(B, \omega^\bullet)) = \mathrm{RHom}_A(N, \omega^\bullet).$$

Thus properties (1) and (2) in Definition 6.1 follow. For property (3),

$$\mathrm{RHom}_B(\omega^\bullet_B, \omega^\bullet_B) = \mathrm{RHom}_A(\omega^\bullet_B, \omega^\bullet) = B,$$

by biduality for the finite A-module B.

For the normalized statement, we have

$$\mathrm{RHom}_B(l, \omega^\bullet_B) = \mathrm{RHom}_A(l, \omega^\bullet),$$

where l is the residue field of B, which is isomorphic to a finite direct sum of k as an A-module. □

8. Dimension function

For a dualizing complex $\omega^\bullet \in \mathrm{D}(A)$ and $\mathfrak{p} \in \operatorname{Spec} A$, define $\delta(\mathfrak{p})$ to be the unique integer d such that

$$\mathrm{H}^{-d} \mathrm{RHom}_{A_\mathfrak{p}}(k(\mathfrak{p}), \omega^\bullet_\mathfrak{p}) \neq 0.$$

This is well-defined, because $\mathrm{RHom}_{A_\mathfrak{p}}(k(\mathfrak{p}), \omega^\bullet_\mathfrak{p})$ is a dualizing complex for the field $k(\mathfrak{p})$, hence isomorphic to a shift of $k(\mathfrak{p})$. If A is local, then $\omega^\bullet$ is normalized if and only if $\delta(\mathfrak{m}) = 0$.

Theorem 8.1. *If $\mathfrak{p} \subseteq \mathfrak{q}$ then*

$$\delta(\mathfrak{p}) = \delta(\mathfrak{q}) + \mathrm{ht}(\mathfrak{q}/\mathfrak{p}).$$

This means that δ is a dimension function.

Proof. Going through a maximal chain of prime ideals between $\mathfrak{p}$ and $\mathfrak{q}$ of length $\mathrm{ht}(\mathfrak{q}/\mathfrak{p})$, we may assume $\mathrm{ht}(\mathfrak{q}/\mathfrak{p}) = 1$. By localization and quotient (Lemma 7.1) we may assume $(A, \mathfrak{m})$ is local, $\mathfrak{q} = \mathfrak{m}$, and $\mathfrak{p} = 0$, so $\dim A = 1$. We may also assume $\omega^\bullet$ is normalized. Then (Theorem 4.2)

$$\widehat{\mathrm{H}^{-1}(\omega^\bullet)} \simeq \mathrm{H}^1_\mathfrak{m}(A)^\vee, \qquad \widehat{\mathrm{H}^0(\omega^\bullet)} \simeq \mathrm{H}^0_\mathfrak{m}(A)^\vee.$$

Since $\mathrm{H}^0_\mathfrak{m}(A)$ is of finite length (being an ideal of A)[2], the finitely generated module $\mathrm{H}^0(\omega^\bullet)$ is also of finite length. Since $\operatorname{Supp}(\omega^\bullet) = \operatorname{Spec} A$ we must have $\operatorname{Supp} \mathrm{H}^{-1}(\omega^\bullet) = \operatorname{Spec} A$, whence by definition $\delta(\mathfrak{p}) = 1$. □

[1] meaning that over a Zariski-locally a shift of A

[2] In fact, since we assumed A is a domain, we have $\mathrm{H}^0_\mathfrak{m}(A) = 0$.

Corollary 8.2. *If A has a dualizing complex then A is catenary,* i.e.

$$\operatorname{ht}(\mathfrak{r}/\mathfrak{q}) + \operatorname{ht}(\mathfrak{q}/\mathfrak{p}) = \operatorname{ht}(\mathfrak{r}/\mathfrak{p})$$

for primes $\mathfrak{r} \supseteq \mathfrak{q} \supseteq \mathfrak{p}$.

9. Support of the dualizing module

Theorem 9.1. *Suppose A is local and let $\omega_A^\bullet$ be a normalized dualizing complex (if exists). Let $d = \dim A$. Then*

$$\operatorname{Ass}_A \mathrm{H}^{-d}(\omega_A^\bullet) = \{\mathfrak{p} \mid \dim A/\mathfrak{p} = d\}.$$

Proof. For every $\mathfrak{p} \in \operatorname{Spec} A$, $\operatorname{RHom}_A(A/\mathfrak{p}, \omega_A^\bullet)$ is a normalized dualizing complex of $A/\mathfrak{p}$, hence is in $\mathrm{D}^{[-\dim A/\mathfrak{p}, 0]} \subseteq \mathrm{D}^{[-d,0]}$ (Lemma 7.1). If there is an injection

$$A/\mathfrak{p} \hookrightarrow \mathrm{H}^{-d}(\omega_A^\bullet),$$

then

$$\mathrm{H}^{-d} \operatorname{RHom}_A(A/\mathfrak{p}, \omega_A^\bullet) = \operatorname{Hom}_A(A/\mathfrak{p}, \mathrm{H}^{-d}(\omega_A^\bullet)) \neq 0.$$

It follows that $\dim A/\mathfrak{p} \geq d$.

Conversely, if $\dim A/\mathfrak{p} = d$, then $\delta(\mathfrak{p}) = d$ (Theorem 8.1), that is,

$$\left(\mathrm{H}^{-d} \operatorname{RHom}_A(A/\mathfrak{p}, \omega_A^\bullet)\right)_\mathfrak{p} \neq 0.$$

Hence there is a map

$$\varphi : A/\mathfrak{p} \longrightarrow \mathrm{H}^{-d}(\omega_A^\bullet)$$

which localizes to a nonzero map

$$0 \neq \varphi_\mathfrak{p} : k(\mathfrak{p}) \longrightarrow \mathrm{H}^{-d}(\omega_A^\bullet)_\mathfrak{p},$$

and so φ is injective. □

Corollary 9.2.

$$\operatorname{Supp} \mathrm{H}^{-d}(\omega_A^\bullet) = \bigcup_{\substack{\mathfrak{p} \in \operatorname{Spec} A \\ \dim A/\mathfrak{p} = d}} V(\mathfrak{p}).$$

The following statements are left as exercises.

$$\operatorname{Ass}_A \mathrm{H}^{-j}(\omega_A^\bullet) \supseteq \{\mathfrak{p} \in \operatorname{Min} A \mid \dim A/\mathfrak{p} = j\} \tag{2}$$

$$\dim \operatorname{Supp} \mathrm{H}^{-j}(\omega_A^\bullet) \leq j \tag{3}$$

$$\mathrm{H}^{-d}(\omega_A^\bullet) = \mathrm{H}^{-d}(\omega^\bullet_{A/\mathrm{H}^0_\mathfrak{m}(A)}), \text{ if } d > 0 \tag{4}$$

Here $\operatorname{Min} A$ is the set of minimal primes of A.

10. Review: dualizing complexes and Cohen–Macaulay rings

For an arbitrary Noetherian local ring $(A, \mathfrak{m}, k)$, let $\omega_{\widehat{A}}^\bullet$ denote the normalized dualizing complex of $\widehat{A}$. Recall that, for $K \in \mathrm{D}^+_{\mathrm{coh}}(A)$, the following are equivalent:

(*i*) $K \otimes^L_A \widehat{A} \simeq \omega_{\widehat{A}}^\bullet[\delta]$ for some $\delta \in \mathbb{Z}$;
(*ii*) $\operatorname{RHom}_A(k, K) \simeq k[\delta]$ for some $\delta \in \mathbb{Z}$;
(*iii*) K has finite injective dimension and the canonical map $A \longrightarrow \operatorname{RHom}_A(K, K)$ is an isomorphism.

If A has a normalized dualizing complex $\omega_A^\bullet$ (for example, if A is complete), put $d = \dim A$ and $\omega_A = \mathrm{H}^{-d}(\omega_A^\bullet)$. Then

$$\omega_A^\bullet \in \mathrm{D}^{\geq -d}(A), \qquad \operatorname{Ass}_A \omega_A = \{\mathfrak{p} \in \operatorname{Spec} A \mid \dim A/\mathfrak{p} = d\} \neq \varnothing.$$

Definition 10.1. A Noetherian local ring A is *Gorenstein* if

$$\omega_{\widehat{A}}^{\bullet} \simeq \widehat{A}[\dim A].$$

Equivalently, $\widehat{A}$ has finite injective dimension over $\widehat{A}$; equivalently, A has finite injective dimension over A (that is, A is a dualizing complex for itself).

Definition 10.2. A Noetherian local ring A is *Cohen–Macaulay* if

$$\omega_{\widehat{A}}^{\bullet} \in \mathrm{D}^{[-\dim A, -\dim A]}(\widehat{A}).$$

This definition does not require a dualizing complex to be defined over A. A finite A-module $M \in \mathrm{D}^{\heartsuit}_{\mathrm{coh}}(A)$ is *Cohen–Macaulay* if

$$\mathrm{RHom}_{\widehat{A}}(\widehat{M}, \omega_{\widehat{A}}^{\bullet})$$

is concentrated in a single degree.

Theorem 10.3. *Let A be a Noetherian local ring of dimension d.*

(*i*) *If A is Cohen–Macaulay, then A is formally unmixed, that is, $\dim \widehat{A}/\mathfrak{p} = d$ for every $\mathfrak{p} \in \mathrm{Ass}_{\widehat{A}} \widehat{A}$.*

(*ii*) *If A is Cohen–Macaulay and $f \in \mathfrak{m}$ is a nonzerodivisor on A, then A/fA is Cohen–Macaulay.*

(*iii*) *Conversely, if $f \in \mathfrak{m}$ is a nonzerodivisor on A and A/fA is Cohen–Macaulay, then A is Cohen–Macaulay.*

Proof. We may assume A is complete. In particular, $\omega_A^{\bullet}$ exists; write $\omega_A = \mathrm{H}^{-d}(\omega_A^{\bullet})$.

For (i), since A is Cohen–Macaulay, biduality gives

$$A \simeq \mathrm{RHom}_A(\omega_A^{\bullet}, \omega_A^{\bullet}) \simeq \mathrm{RHom}_A(\omega_A, \omega_A).$$

Taking H^0 yields $A \simeq \mathrm{Hom}_A(\omega_A, \omega_A)$. For arbitrary modules M, N, a nonzerodivisor on N is automatically a nonzerodivisor on $\mathrm{Hom}_A(M, N)$. The set of zerodivisors of ω_A being the union of $\mathrm{Ass}_A \omega_A = \{\mathfrak{p} \in \mathrm{Spec}\, A \mid \dim A/\mathfrak{p} = d\}$, showing that $\mathrm{Ass}_A A \subseteq \{\mathfrak{p} \in \mathrm{Spec}\, A \mid \dim A/\mathfrak{p} = d\}$, so every associated prime of A has quotient dimension d. Thus A is unmixed.

For (ii) and (iii), use the distinguished triangle

$$A \xrightarrow{\times f} A \longrightarrow A/fA \longrightarrow +1.$$

It gives

$$\omega_{A/fA}^{\bullet} := \mathrm{RHom}_A(A/fA, \omega_A^{\bullet}) \simeq (\omega_A^{\bullet} \otimes_A^L A/fA)[-1],$$

where the left hand side is the normalized dualizing complex of A/fA.

For (ii), f lies in no minimal prime of A. By the description of $\mathrm{Ass}_A \omega_A$, it therefore lies in no associated prime of ω_A, so multiplication by f on ω_A is injective. Since $\omega_A^{\bullet} \simeq \omega_A[d]$, the displayed formula becomes

$$\omega_{A/fA}^{\bullet} \simeq (\omega_A/f\omega_A)[d-1].$$

It is concentrated in degree $-(d-1)$, as required.

For (iii), $\dim A/fA = d - 1$, so the same formula and the Cohen–Macaulay hypothesis show that $\omega_A^{\bullet} \otimes_A^L A/fA$ is concentrated in degree $-d$. Nakayama (Corollary 1.15) gives $\omega_A^{\bullet} \in \mathrm{D}^{\leq -d}(A)$. Together with $\omega_A^{\bullet} \in \mathrm{D}^{\geq -d}(A)$, this proves A is Cohen–Macaulay. □

11. Cohomological depth

Let $(A, \mathfrak{m}, k)$ be a Noetherian local ring and $M \neq 0$ a finite A-module.

Definition 11.1. Set

$$\operatorname{depth} M := \min\{j \mid \operatorname{Ext}^j_A(k, M) \neq 0\}.$$

Proposition 11.2. *One has*

$$\operatorname{depth} M = \min\{j \mid \mathrm{H}^j_{\mathfrak{m}}(M) \neq 0\}.$$

If A has a normalized dualizing complex $\omega^\bullet_A$ and we set $\mathbb{D}(M) = \operatorname{RHom}_A(M, \omega^\bullet_A)$, then

$$\operatorname{depth} M = -\max\{j \mid \mathrm{H}^j\mathbb{D}(M) \neq 0\}.$$

In particular, $\operatorname{depth} M = \operatorname{depth} \widehat{M}$*, and depth is well-defined (*i.e. *has finite value).*

Proof. Since the completion is flat with residue field k,

$$\operatorname{RHom}_A(k, M) \otimes_A \widehat{A} \simeq \operatorname{RHom}_{\widehat{A}}(k, \widehat{M}).$$

Faithful flatness shows that completion preserves the first nonzero cohomology degree, so $\operatorname{depth} M = \operatorname{depth} \widehat{M}$. Similarly, we may assume A is complete, hence $M \in \mathrm{D}_{\mathfrak{m}\text{-comp}}(A)$.

Since k is $\mathfrak{m}$-torsion we have

$$\operatorname{RHom}_A(k, R\Gamma_{\mathfrak{m}} M) \simeq \operatorname{RHom}_A(k, M)$$

The dual Nakayama (Lemma 5.1) applied to $R\Gamma_{\mathfrak{m}} M$ identifies this degree with the first nonzero degree of $R\Gamma_{\mathfrak{m}} M$. Thus

$$\operatorname{depth} M = \min\{j \mid \mathrm{H}^j \operatorname{RHom}_A(k, R\Gamma_{\mathfrak{m}} M) \neq 0\} = \min\{j \mid \mathrm{H}^j_{\mathfrak{m}}(M) \neq 0\}.$$

This shows that $\operatorname{depth} M$ is well-defined, since if $R\Gamma_{\mathfrak{m}}(M) = 0$, then $M = \widehat{R\Gamma_{\mathfrak{m}}(M)} = 0$.

Second, local duality gives

$$\widehat{\mathbb{D}(M)} \simeq \operatorname{RHom}_A(R\Gamma_{\mathfrak{m}} M, E).$$

Note that E is injective, and every $\mathfrak{m}^\infty$-torsion module $X \neq 0$ satisfies $\operatorname{RHom}_A(X, E) \neq 0$, because X contains a copy of k. So the first nonzero degree on the right corresponds to the negative of the last nonzero degree of $\mathbb{D}(M)$. Completion is faithfully flat, which proves the second equality. □

Corollary 11.3. *A is Cohen–Macaulay if and only if* $\operatorname{depth} A = \dim A$.

Proposition 11.4 (Classical description of depth)**.** *For a nonzero finite A-module M,* $\operatorname{depth} M$ *is the maximum length of an M-regular sequence in $\mathfrak{m}$.*

Proof. Temporarily denote the maximum length by $r(M)$, and induct on $r(M)$. The following two observations are also useful independently.

(*i*) If $r(M) = 0$, then $\operatorname{depth} M = 0$. Indeed, every element of $\mathfrak{m}$ is a zerodivisor on M. The theory of associated primes and prime avoidance then give $\mathfrak{m} \in \operatorname{Ass}_A M$, equivalently $\operatorname{Hom}_A(k, M) \neq 0$.

(*ii*) If $r(M) > 0$ and $f \in \mathfrak{m}$ is an M-nonzerodivisor, then

$$\operatorname{depth}(M/fM) = \operatorname{depth} M - 1. \tag{5}$$

In particular, this shows $r(M) \leq \operatorname{depth} M < \infty$, so the induction is valid. To see (5), apply $\operatorname{RHom}_A(k, -)$ to $M \xrightarrow{f} M \longrightarrow M/fM \longrightarrow +1$. Multiplication by f is zero on each $\operatorname{Ext}^j_A(k, M)$, since these are k-vector spaces. The resulting long exact sequence gives

$$0 \longrightarrow \operatorname{Ext}^j_A(k, M) \longrightarrow \operatorname{Ext}^j_A(k, M/fM) \longrightarrow \operatorname{Ext}^{j+1}_A(k, M) \longrightarrow 0.$$

Also $\operatorname{Hom}_A(k, M) = 0$ because f is injective on M; hence the first nonvanishing degree drops by one.

One has $r(M/fM) < r(M)$: a regular sequence on M/fM lifts to a regular sequence on M after prepending f. Applying the induction hypothesis to M/fM proves that $r(M) \geq r(M/fM) + 1 = \operatorname{depth} M$, as desired. □

Corollary 11.5. *A is Cohen–Macaulay if and only if there is a regular sequence generating an $\mathfrak{m}$-primary ideal. This implies that there exists a nonzero finite-length A-module of finite projective dimension.*

12. The Auslander–Buchsbaum formula

An important tool to study modules over a Noetherian local ring is minimal free resolutions. I have asked GPT to generate a short introduction to it, which is more than enough for the Auslander–Buchsbaum formula.

12.1. **Minimal free resolutions.** Let A be a ring. Recall that an object $M \in \mathrm{D}(A)$ is *pseudo-coherent* if it is represented by a bounded above complex of finite projective A-modules. Over a Noetherian ring, this is equivalent to $M \in \mathrm{D}^-_{\mathrm{coh}}(A)$.

Now suppose $(A, \mathfrak{m}, k)$ is local. A bounded above complex $F^\bullet$ of finite free A-modules is *minimal* if

$$d^i(F^i) \subseteq \mathfrak{m}F^{i+1} \qquad \text{for every } i.$$

Proposition 12.1. *Every pseudo-coherent object $M \in \mathrm{D}(A)$ admits a minimal free resolution $F^\bullet \xrightarrow{\sim} M$. This resolution is unique up to an isomorphism of complexes, and*

$$\operatorname{rank}_A F^i = \dim_k \mathrm{H}^i(k \otimes^L_A M).$$

Moreover, M is perfect if and only if $F^\bullet$ is bounded.

Proof. Choose a bounded above finite free resolution of M. If a differential has a unit entry, changes of bases split off a contractible summand

$$[\, A \xrightarrow{1} A \,].$$

Starting with the highest nonzero degree and proceeding downwards, remove these summands. Only finitely many cancellations affect each degree, and the remaining complex $F^\bullet$ is minimal.

The differentials of $k \otimes_A F^\bullet$ are zero, so

$$\mathrm{H}^i(k \otimes^L_A M) \simeq k \otimes_A F^i,$$

which proves the rank formula. Any two bounded above projective resolutions of M are homotopy equivalent. After tensoring with k, a homotopy equivalence between minimal resolutions is an isomorphism in each degree, because their differentials vanish. Nakayama's lemma then shows that the homotopy equivalence itself is an isomorphism in each degree.

If $F^\bullet$ is bounded, then M is perfect. Conversely, if M is perfect, then $k \otimes^L_A M$ has bounded cohomology, so the rank formula forces $F^\bullet$ to be bounded. □

In particular, for a finite A-module M placed in degree zero, its minimal free resolution has the form

$$\cdots \longrightarrow A^{\oplus c_2} \longrightarrow A^{\oplus c_1} \longrightarrow A^{\oplus c_0} \longrightarrow M \longrightarrow 0, \qquad c_j = \dim_k \operatorname{Tor}^A_j(k, M).$$

All differential matrices have entries in $\mathfrak{m}$. If $M \neq 0$ has finite projective dimension p, then $c_p \neq 0$ and $c_j = 0$ for $j > p$.

12.2. **The Auslander–Buchsbaum formula.**

Theorem 12.2. *Let $M \in \mathrm{D}^{\heartsuit}_{\mathrm{coh}}(A)$ be nonzero with $\operatorname{pd}_A M < \infty$. Then*

$$\operatorname{pd}_A M + \operatorname{depth} M = \operatorname{depth} A.$$

Proof. Take a minimal finite free resolution, with $p = \operatorname{pd}_A M$,

$$\left[A^{\oplus c_p} \xrightarrow{\varphi} A^{\oplus c_{p-1}} \longrightarrow \cdots \longrightarrow A^{\oplus c_0}\right] \simeq M, \qquad c_j = \dim_k \operatorname{Tor}^A_j(k, M).$$

Put $F = A^{\oplus c_p}$, $G = A^{\oplus c_{p-1}}$, and let $Y = \sigma_{\geq -p+2} M^\bullet$ and $X = \sigma_{\leq -p+1} M^\bullet$ be the indicated brutal truncations of this resolution. Thus $Y \in \mathrm{D}^{\geq -p+2}(A)$, and there are exact triangles

$$Y \longrightarrow M \longrightarrow X \longrightarrow +1, \qquad G[p-1] \longrightarrow X \longrightarrow F[p] \longrightarrow +1.$$

For $p = 0$ the result is trivial because M is finite free, so assume $p > 0$; for $p = 1$ take $Y = 0$. Write $j_0 = \operatorname{depth} A - p$. Since every term of the resolution is finite free,

$$\operatorname{Ext}^j_A(k, M) = 0 \qquad (j < j_0).$$

For $j = j_0$, the bound on Y gives $\operatorname{Ext}^{j_0}_A(k, Y) = 0 = \operatorname{Ext}^{j_0+1}_A(k, Y)$, so

$$\operatorname{Ext}^{j_0}_A(k, M) \simeq \operatorname{Ext}^{j_0}_A(k, X).$$

The second triangle gives the exact sequence

$$0 \longrightarrow \operatorname{Ext}^{j_0}_A(k, X) \longrightarrow \operatorname{Ext}^{\operatorname{depth} A}_A(k, F) \xrightarrow{\operatorname{Ext}^{\operatorname{depth} A}_A(k, \varphi)} \operatorname{Ext}^{\operatorname{depth} A}_A(k, G).$$

Minimality means that all entries of φ lie in $\mathfrak{m}$, or that $\varphi \in \mathfrak{m} \operatorname{Hom}_A(F, G)$. Since $\mathfrak{m}$ annihilates $\operatorname{Ext}^{\operatorname{depth} A}_A(k, A)$, the last map is zero. Moreover, $F \neq 0$, so $\operatorname{Ext}^{\operatorname{depth} A}_A(k, F) \neq 0$. Consequently $\operatorname{Ext}^{j_0}_A(k, M) \neq 0$, and $\operatorname{depth} M = j_0$, as desired. □

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